

```

import pandas as pd
import numpy as np

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

# -----
# Create Sample Application Dataset
# -----

app = pd.DataFrame({
    "ID": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "CODE_GENDER": ["M", "F", "M", "F", "M", "F", "M", "F", "M", "F"],
    "FLAG_OWN_CAR": ["Y", "N", "Y", "N", "Y", "N", "Y", "N", "Y", "N"],
    "FLAG_OWN_REALTY": ["Y", "Y", "N", "Y", "N", "Y", "Y", "N", "Y", "N"],
    "CNT_CHILDREN": [0, 1, 0, 2, 1, 0, 0, 1, 2, 0],
    "AMT_INCOME_TOTAL": [250000, 180000, 300000, 200000, 350000, 150000, 400000, 220000, 270000, 190000],
    "NAME_INCOME_TYPE": [
        "Working", "Working", "Commercial associate", "Pensioner",
        "Working", "State servant", "Working", "Student",
        "Commercial associate", "Working"
    ]
})

# -----
# Create Sample Credit Dataset
# -----

credit = pd.DataFrame({
    "ID": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "STATUS": ["0", "1", "C", "0", "2", "X", "0", "3", "C", "1"]
})

# -----
# Convert STATUS into Target
# 0 = Approved
# 1 = Rejected
# -----

credit["STATUS"] = credit["STATUS"].replace({
    "C": 0,
    "X": 0,
    "0": 0,
    "1": 1,
    "2": 1,
    "3": 1,
    "4": 1,
    "5": 1
})

credit.rename(columns={"STATUS": "Target"}, inplace=True)

# -----
# Encode Categorical Columns
# -----

encoder = LabelEncoder()

for col in app.columns:
    if app[col].dtype == object:
        app[col] = encoder.fit_transform(app[col])

# -----
# Merge Data
# -----

data = pd.merge(app, credit, on="ID")

print(data.head())

# -----
# Prepare Features
# -----

```

```

X = data.drop(["ID","Target"], axis=1)
y = data["Target"]

# -----
# Scale Data
# -----

scaler = StandardScaler()

X = scaler.fit_transform(X)

# -----
# Split Dataset
# -----

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
)

# -----
# Train Model
# -----

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)

# -----
# Prediction
# -----

prediction = model.predict(X_test)

print("\nAccuracy:", accuracy_score(y_test, prediction))
print("\nClassification Report")
print(classification_report(y_test, prediction))

```

	ID	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	\
0	1	1	1	1	0	
1	2	0	0	1	1	
2	3	1	1	0	0	
3	4	0	0	1	2	
4	5	1	1	0	1	

	AMT_INCOME_TOTAL	NAME_INCOME_TYPE	Target
0	250000	4	0
1	180000	4	1
2	300000	0	0
3	200000	1	0
4	350000	4	1

Accuracy: 1.0

Classification Report				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	2
1	1.00	1.00	1.00	1
accuracy			1.00	3
macro avg	1.00	1.00	1.00	3
weighted avg	1.00	1.00	1.00	3

```

/tmp/ipykernel_3809/4065840017.py:42: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a fu
credit["STATUS"] = credit["STATUS"].replace({

```

